Theorem 1. *For $0 < \alpha < 1$, let $p_\alpha(j)$ the pmf of sibuya distribution:*

$$p_\alpha(j) := \frac{\alpha \prod_{i=1}^{j-1}(i-\alpha)}{j!} = \alpha \frac{\Gamma(j-\alpha)}{\Gamma(j+1)\Gamma(1-\alpha)}, \qquad j \geq 1.$$

Then it holds that

$$\mathrm{fisher}(\alpha) := \alpha^{-2} + \sum_{j=1}^{\infty} p_\alpha(j) \frac{1}{\alpha(j-\alpha)} = \frac{\pi}{\alpha \sin(\pi\alpha)}.$$

Using $\sin(\pi\alpha) < \pi(1-\alpha)$, we know that the fisher information is larger than $\frac{1}{\alpha(1-\alpha)}$.

Proof. Using

$$\frac{1}{j-\alpha} = \int_0^1 x^{j-\alpha-1}\, dx, \qquad j \geq 1,$$

we obtain

$$\sum_{j=1}^{\infty} \frac{p_\alpha(j)}{j-\alpha} = \int_0^1 x^{-\alpha-1} \sum_{j=1}^{\infty} p_\alpha(j) x^j \, dx.$$

Now the probability generating function of the Sibuya distribution is

$$\sum_{j=1}^{\infty} p_\alpha(j) x^j = 1 - (1-x)^\alpha, \qquad x \in [0,1].$$

Therefore,

$$\sum_{j=1}^{\infty} \frac{p_\alpha(j)}{j-\alpha} = \int_0^1 x^{-\alpha-1} \big(1 - (1-x)^\alpha\big)\, dx.$$

Hence

$$\mathrm{fisher}(\alpha) = \alpha^{-2} + \alpha^{-1} \int_0^1 x^{-\alpha-1} \big(1 - (1-x)^\alpha\big)\, dx.$$

Using integration by parts,

$$\begin{aligned}
\int_0^1 x^{-\alpha-1} \big(1 - (1-x)^\alpha\big)\, dx &= \left[-\frac{1}{\alpha} x^{-\alpha} \big(1 - (1-x)^\alpha\big) \right]_0^1 + \int_0^1 x^{-\alpha} (1-x)^{\alpha-1}\, dx \\
&= -\frac{1}{\alpha} + B(1-\alpha, \alpha).
\end{aligned}$$

where B is the Beta function. Since

$$B(1-\alpha, \alpha) = \Gamma(1-\alpha)\Gamma(\alpha),$$

it follows that

$$\mathrm{fisher}(\alpha) = \alpha^{-2} + \alpha^{-1} \left(-\alpha^{-1} + \Gamma(1-\alpha)\Gamma(\alpha) \right) = \frac{\Gamma(1-\alpha)\Gamma(\alpha)}{\alpha}.$$

Finally, Euler's reflection formula gives

$$\Gamma(\alpha)\Gamma(1-\alpha) = \frac{\pi}{\sin(\pi\alpha)},$$

and thus

$$\mathrm{fisher}(\alpha) = \frac{\pi}{\alpha \sin(\pi\alpha)}.$$

$\square$

Let $k_{n,j}$ be the emperical frequency generated by the Gibbs partition and let $P_{n,j} = \frac{k_{n,j}}{k_n}$. We know that the pointwise convergence hold

$$P_{n,j} \to^p p_\alpha(j).$$

As a corollary, one can show

$$\sum_j |P_{n,j} - p_\alpha(j)| \to 0. \tag{1}$$

We aim to show the asymptotic normality of the QMLE under a certain assumption on the distributional convergence of the random measure $\sqrt{K_n}(P_{n,\cdot} - p_\alpha)$ on $\mathbb{N}$. Recall that the QMLE is defined as the root of the system

$$\Psi_n(x) = 0$$

where Ψ_n is the random function given by

$$\Psi_n(x) = \frac{1}{k_n}\Big(\frac{k_n - 1}{x} - \sum_{j=2}^{n} k_{n,j} \sum_{i=1}^{j-1} \frac{1}{i-x}\Big) = -\frac{1}{k_n} + \frac{1}{x} - \sum_{j=2}^{n} P_{n,j} \sum_{i=1}^{j-1} \frac{1}{i-x}$$

(Ψ_n is the derivative of the log likelihood of the Ewens–Pitman with $\theta = 0$ normalized by k_n). We also define the deterministic function Ψ as

$$\Psi(x) = \frac{1}{\alpha} - \sum_{j=2}^{\infty} p_\alpha(j) \frac{1}{i-x}.$$

This is obtained from Ψ_n with $P_{n,j}$ replaced by $p_\alpha(j)$ and ignoring the diminishing term $1/k_n$. We showed in previous paper that the map $\Psi : (0,1) \to \mathbb{R}$ satisfies

- $\Psi : (0,1) \to \mathbb{R}$ is C^1.
- Ψ is strictly decreasing and

$$\Psi'(x) = -\frac{1}{x^2} - \sum_{j=2}^{\infty} p_\alpha(j) \sum_{i=1}^{j-1} \frac{1}{(i-x)^2} < 0$$

- $\Psi'(\alpha) = -i(\alpha) < 0$, where $i(\alpha)$ is the Fisher Information of the discrete distribution $p_\alpha(j)$.
- Ψ has its unique root at $x = \alpha$, i.e., $\Psi(\alpha) = 0$.
- $\Psi_n'(x)$

Furthermore, using (1), one can show that for any closed interval $I = [s,t]$ with $0 < s < t < 1$,

$$\sup_{x \in I} |\Psi_n(x)' - \Psi(x)'| \to^p 0.$$

Now, using $\hat{\Psi}_n(\hat{\alpha}_n) = \Psi(\alpha) = 0$, by Taylor's expansion,

$$\Psi_n(\alpha) - \Psi(\alpha) = \Psi_n(\alpha) - \Psi_n(\hat{\alpha}_n) = \Psi_n'(\tilde{\alpha}_n)(\alpha - \hat{\alpha}_n).$$

Now we assume that for certain test function $f : \mathbb{N} \to \mathbb{R}$,

$$\sqrt{K_n}\Big(\sum_j f(j)(P_{n,j} - p_\alpha(j))\Big) \to N(0, f^\top \Gamma_\alpha f)$$

where $\Gamma_\alpha = D(p_\alpha) - p_\alpha p_\alpha^\top$

Now, using your L_2 distributional convergence, taking test function $f_x : \mathbb{N} \to \mathbb{R}$ as $f_x(j) = \sum_{i=1}^{j-1} \frac{1}{i-x}$ for each x, we would get

$$\forall x \in (0,1), \quad \sqrt{K_n}(\Psi_n(x) - \Psi(x)) \to N(0, h_x^\top \Gamma_\alpha h_x)$$

This in particular $\Psi_n(x) \to^p \Psi(x)$ pointwisely and hence we have the consistency of the QMLE.

Inportantly, taking $x = \alpha$. By the same argument, now taking the test function $h_x : \mathbb{N} \to \mathbb{R}$ as $h_x(j) = \sum_{i=1}^{j-1} \frac{1}{(i-x)^2}$ for each x, we also get

$$\forall x \in (0,1), \quad \Psi_n'(x) \to^p \Psi'(x).$$